

IIE ROSEBANK INTERNATIONAL
XISD5319 — Work Integrated Learning

TASK 3

FINAL PROJECT REPORT

COSUP Website & Mobile Application
Community Oriented Substance Use Programme

Prepared By	The 5 Watermelons
Team Members	Zandile Selao Tshiamo Moseitlha Matlhogonolo Keebine Muhluri Nkuna Ndabezinhle Mathunyane
Module	XISD5319 — Work Integrated Learning
Institution	IIE Rosebank International, Pretoria CBD
Client	Community Oriented Substance Use Programme (COSUP)
Date	June 2026
Version	3.0 — Final Submission

GitHub Prototype: <https://tmosetlha.github.io/cosup-prototype/>
Website V1: https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/

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1. Introduction

1.1 Purpose of This Report

This document constitutes the Task 3 Final Project Report for the COSUP Website and Mobile Application, submitted in fulfilment of the XISD5319 Work Integrated Learning module at IIE Rosebank International, Pretoria CBD. It consolidates all project deliverables into a single, comprehensive submission encompassing the revised project plan, revised requirement analysis, revised system design, prototype screenshots, working system links, and the team's PowerPoint presentation slides.

The system developed is a dual-platform digital solution — a responsive website and an Android mobile application — built for the Community Oriented Substance Use Programme (COSUP), a harm-reduction public health initiative established in 2015 through a partnership between the City of Tshwane and the University of Pretoria's Department of Family Medicine.

1.2 Project Background

COSUP has operated for over a decade, supporting more than 5 000 individuals across 16 active sites in the Tshwane metropolitan area, without any dedicated digital presence. The 5 Watermelons were commissioned through the WIL module to address this gap by delivering two integrated platforms that provide an accessible, confidential, and multilingual gateway to COSUP's life-saving services.

Both platforms were developed in full compliance with the Protection of Personal Information Act (POPIA, Act 4 of 2013). No sensitive health data is stored or processed on either platform. The system functions strictly as a public-facing information, navigation, and communication tool.

1.3 Team

Team Member	Role	Student Number
Zandile Selao	Project Manager	ST10436981
Tshiamo Mosetlha	Lead Developer	(TM_UI)
Matlhogonolo Keebine	UI/UX Designer	ST10452412
Muhluri Nkuna	Business Analyst	ST10437226
Ndabezinhle Mathunyane	Developer / QA Tester	ST10457068

2. Revised Project Plan

2.1 Executive Summary of Changes

The original project plan, submitted as Task 1 in March 2026, established five sequential phases across a timeline concluding on 13 May 2026. Following the Task 1 submission, the team conducted a structured retrospective to assess delivery accuracy, scope adherence, and schedule alignment. The revised plan reflects updates to the project timeline, role clarifications following team workload rebalancing, and an accurate record of deliverables completed versus originally planned.

Key changes in this revision include: (1) the extension of the final submission milestone to accommodate additional UAT cycles with COSUP; (2) an updated risk register reflecting risks that materialised during development; and (3) the addition of a working system link for Website Version 1, which was not present in the original charter.

2.2 Revised Project Objectives

The following objectives were set at project inception and are reviewed here against their delivery status:

Ref	Objective	Status
O1	Develop a professional responsive website and Android mobile application accurately representing COSUP.	Delivered
O2	Implement full multilingual interface support across seven South African languages.	Delivered — language toggle functional on website
O3	Create an accessible clinic and service point locator integrated with the maps API.	Delivered — 16 sites mapped with address and hours
O4	Establish a direct crisis support channel (Hopeline) on the mobile application.	Delivered — one-tap call and WhatsApp button implemented
O5	Ensure full POPIA compliance across both platforms.	Delivered — no sensitive health data stored; consent on all forms
O6	Deliver a complete, documented handover package to COSUP.	In Progress — source code and documentation being finalised

2.3 Revised Project Phases and Schedule

The table below presents the revised project schedule, updated to reflect actual completion dates and outstanding deliverables as at June 2026.

#	Phase	Deliverable	Original Deadline	Actual / Revised Status
1	Organisation Analysis	Background research; stakeholder mapping	Feb 2026	Complete
2	UI/UX Prototype	High-fidelity prototype; COSUP feedback	Mar 2026	Complete — Deployed
3	Project Plan (Task 1)	Project Charter; WBS; Gantt; PERT	23 Apr 2026	Complete — Submitted
3B	Requirement Analysis & System Design (Task 2)	UML; Class Diagrams; DB Design; ERD; Architecture	May 2026	Complete — Submitted
4	Full Development & Testing	Website V1; Android app; CRUD; DB integration; multilingual; UAT	May 2026	Complete — Systems Functional
5	Final Submission (Task 3)	Final Report; Presentation; Source code handover	22 Jun 2026	In Progress — Current Document

2.4 Revised Roles and Responsibilities

The original role assignments remained largely intact throughout the project. The following refinements were documented during the final phase:

Team Member	Role	Key Contributions — Final Phase
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Zandile Selao	Project Manager	Coordinated final submission; managed COSUP communications; compiled final documentation package.
Tshiamo Mosetlha	Lead Developer	Led Android Kotlin development; integrated OSMDroid maps; built dark/light mode toggle; managed GitHub deployment of Website V1.
Matlhogonolo Keebine	UI/UX Designer	Implemented website frontend (HTML/CSS/JS); refined multilingual toggle; ensured responsive design across all pages.
Muhluri Nkuna	Business Analyst	Developed ASP.NET MVC backend; designed SQL Server database; implemented REST API endpoints; POPIA consent module.
Ndabezinhle Mathunyane	Dev / QA Tester	Executed full test plan; cross-browser and cross-device testing; compiled UAT results; finalised Maintenance and Handover Guide.

2.5 Revised Risk Register

The following risks from the original register are updated to reflect their actual outcome during the project lifecycle:

Ref	Risk Description	Original Rating	Outcome
R1	Multilingual integration complexity causing delays.	High	Materialised — resolved using i18n JSON resource files. Language toggle functional across all 7 languages.
R4	Team member unavailability.	High	Partially materialised — workload rebalanced across remaining members without impacting final delivery.
R6	COSUP point of contact unresponsive.	High	Materialised — COSUP response times were slower than anticipated. Team proceeded with prototype sign-off and continued independently.

R8	COSUP branding assets delayed.	High	Materialised — team sourced approved brand colours and logo from COSUP's public materials to proceed.
R9	Development phase exceeding estimated duration.	High	Managed — Gantt was revised internally; final deadline met.
R10	POPIA compliance breach.	Critical	Did not materialise — no sensitive data stored. Consent mechanism implemented on all data capture forms.

2.6 Revised Budget Summary

The project was delivered within its self-funded budget. No additional monetary expenditure beyond the original estimates was incurred. All development tooling remained at zero cost through student and open-source licensing. The estimated commercial value of the delivered system remains conservatively assessed at between R265 000 and R505 000.

3. Revised Requirement Analysis

3.1 Summary of Changes from Task 2

The Requirement Analysis Document submitted in Task 2 defined 20 functional requirements and a full UML Use Case Diagram. During the development phase, three functional requirements were refined to better reflect actual system behaviour as built. No requirements were removed from scope. One additional use case was identified during UAT.

Ref	Original Requirement	Change Type	Revised Requirement
FR7	Content Management: admin interface for authorised staff.	Refined	Admin interface implemented with role-based access control via JWT authentication. Admins can update clinic hours, publish news, and view referral submissions.
FR8	Self-referral form with POPIA-compliant consent.	Refined	Self-referral form accepts anonymous submissions (userId nullable). POPIA consent checkbox is mandatory. Case worker is notified via email on submission.
UC-NEW	— (New use case identified in UAT)	Addition	UC21: Dark Mode Toggle — Users may switch between dark and light mode on the website. Language preference and theme preference are stored in the User table.

3.2 Revised Functional Requirements

The complete functional requirements are listed below, incorporating all revisions identified during development and UAT. Requirements unchanged from Task 2 are included for completeness.

Ref	Actor	Requirement
FR1	Visitor / Public	The system must present clear information about COSUP's mission, history, partnerships, and impact statistics.
FR2	Visitor / Public	The system must list all six core COSUP services with descriptions: OAT, NSP, Psychosocial Support, Vocational Training, HIV & HCV Prevention, and Community Integration.
FR3	All Users	The system must allow users to switch the interface language among seven identified South African languages (English, IsiZulu, Xitsonga, TshiVenda, SeTswana/Sepedi, Afrikaans, Xhosa).
FR4	Visitor / Public	The system must allow users to search for and view nearby COSUP clinics and outreach points displaying address, operating hours, and phone number.
FR5	Person in Crisis	The mobile application must provide a consistently visible, one-tap button to immediately call or WhatsApp the COSUP Hopeline (0800 611 197).
FR6	Visitor / Public	The system must display a news and updates section fed from the NewsItem database table.
FR7 (Revised)	COSUP Admin Staff	The system must include a secure, role-based admin interface (JWT-protected) for authorised COSUP staff to update clinic hours, publish news items, and view self-referral submissions.
FR8 (Revised)	Prospective Client	The system must include a POPIA-compliant self-referral form. Anonymous submissions are permitted (userId nullable). A mandatory consent checkbox must be ticked before submission. The submitting case worker receives an email notification.

FR9	Prospective Client	The system must support user registration with name, email, phone, password (BCrypt-hashed), and language preference.
FR10	Registered User	The system must support user login and JWT-authenticated session management.
FR11	Registered User	The system must send an email confirmation upon successful registration via the configured email service provider.
FR12	Mobile App User	The mobile application must display an interactive map with all 16 COSUP site pins using the Google Maps Android SDK / OSMDroid.
FR13	Mobile App User	The mobile application must display the services directory and news feed.
FR14	COSUP Admin Staff	The system must log all admin actions (create/update/delete) to the AuditLog table.
FR15	COSUP Management	The system must generate printable PDF reports of referral submissions, site listings, and Hopeline usage statistics.
UC21 (New)	All Users — Website	The system must support a dark mode / light mode toggle on the website. The selected theme must persist in the user's browser session.

3.3 Non-Functional Requirements — Revised

Non-functional requirements remain substantially unchanged from Task 2. The following refinements were documented during the final development phase:

Ref	Category	Requirement
NFR1	Compatibility	The mobile app targets Android (API 24+). The website must be fully responsive across all modern desktop and mobile browsers (Chrome, Firefox, Safari, Edge).
NFR2	Performance	All pages must load within 3 seconds on standard broadband or 4G connections. The clinic map must render all 16 pins within 2 seconds of page load.
NFR3	Security — Encryption	All data transmission must be secured using HTTPS/TLS encryption. HTTP requests must be redirected to HTTPS.
NFR4	Security — Data Privacy	No sensitive health data (HIV status, HCV status, OAT medication) shall be stored on the client device or backend database under any circumstances.
NFR5	POPIA Compliance	The system must comply fully with POPIA Act 4 of 2013, including user consent on all data capture, and user rights to access, correct, or request deletion of their data.
NFR6	Access Control	All admin functions must be protected by JWT-based role-based access control. Unauthenticated requests to admin endpoints must return HTTP 401.
NFR7	Usability	The interface must be navigable by users with varying digital literacy levels. Key actions (Find Help, Hopeline) must be reachable within two taps or clicks from the home screen.
NFR8	Maintainability	All source code must be commented to a level that enables a third-party developer to maintain the system without prior briefing.
NFR9	No Third-Party Tracking	No third-party analytics or tracking tools shall be embedded without COSUP's explicit prior written approval.
NFR10 (Revised)	Theme	The website must support both dark mode and light mode. The default mode must be dark. The user's preference is stored in sessionStorage for the duration of their visit.

4. Revised System Design

4.1 Summary of Changes from Task 2

The system design documented in Task 2 was implemented as specified, with the following refinements applied during the development phase. These changes reflect decisions made to resolve technical constraints encountered during build and to improve the system's real-world viability within the academic infrastructure available to the team.

Component	Original Design	Revised Design
Maps Integration — Mobile	Google Maps Android SDK	OSMDroid (OpenStreetMap) — selected due to API key cost constraints for academic deployment. Functionally equivalent: 16 site pins, tap-for-detail, user location.
Authentication — Token Storage	JWT stored server-side in Session table	JWT stored in HTTP-only cookie on website; in-memory on mobile app. Session table retained for audit purposes.
Theme Preference	Not in original design	Dark/light mode toggle added to website. Preference stored in sessionStorage (client-side). User table langPref column extended to include theme preference in future versions.
Production Hosting	Azure Web App + Azure SQL	Website V1 deployed to GitHub Pages (static HTML/CSS/JS). ASP.NET MVC backend and SQL Server running on CloudLabs academic environment. Source code and scripts provided for Azure deployment.

4.2 Revised High-Level Architectural Design

The COSUP system retains its three-tier client-server architecture as specified in Task 2. The presentation tier, application/business logic tier, and data persistence tier remain clearly separated, with no direct client-to-database communication.

Tier 1 — Presentation Layer (Client)

Website: HTML5, CSS3, JavaScript — deployed on GitHub Pages (V1) and CloudLabs (full ASP.NET MVC version). Fonts: DM Sans and Playfair Display.

Android Mobile Application: Kotlin, Android Studio — OSMDroid for interactive clinic map; Google Maps Android SDK available for production deployment.

Tier 2 — Application / Business Logic Layer (Server)

ASP.NET MVC, C#, Visual Studio 2022, hosted on IIS / Azure Web App for production. REST API controllers handle all authentication, data retrieval, form processing, and admin operations. External services: Mailgun/SendGrid (email), OSMDroid / Google Maps API (maps), i18n engine (multilingual).

Tier 3 — Data Persistence Layer (Database)

Microsoft SQL Server (Developer Edition) — nine tables: User, Session, Site, SiteService, Service, Referral, NewsItem, HopelineLog, AuditLog. All tables in Third Normal Form (3NF).

4.3 Revised Database Design

The database schema is unchanged from Task 2 with one addition: the User table's langPref column has been clarified to store both the language code (e.g., 'en', 'zu') and in the next release cycle, the theme preference ('dark', 'light'). The nine-table structure remains as specified.

The following table summarises the nine database entities and their primary purpose:

Table	Primary Key	Purpose
User	userId (INT, PK)	Stores registered user accounts including hashed passwords and language preference.
Session	sessionId (INT, PK)	Tracks active JWT sessions per user for audit and security management.
Site	siteId (INT, PK)	Stores all 16 COSUP clinic/service point records including GPS coordinates, address, hours, and phone.
SiteService	siteServiceId (INT, PK)	Junction table resolving the many-to-many relationship between Site and Service.
Service	serviceId (INT, PK)	Stores the six COSUP service types with descriptions and icon paths.
Referral	referralId (INT, PK)	Stores self-referral form submissions. userId is nullable to permit anonymous submissions.
NewsItem	newsId (INT, PK)	Stores published news and announcements managed by COSUP admin staff.
HopelineLog	logId (INT, PK)	Logs anonymous Hopeline button interactions (call vs WhatsApp, web vs mobile) for usage reporting.
AuditLog	logId (INT, PK)	Records all admin actions (updates, publishes, deletes) for accountability and audit trail.

4.4 Revised Technology Stack

Layer	Technology	Notes
Frontend — Website	HTML5, CSS3, JavaScript	DM Sans + Playfair Display fonts. Dark/light mode toggle. Deployed to GitHub Pages (V1) and CloudLabs.
Frontend — Mobile	Kotlin, Android Studio	OSMDroid for maps (replaces Google Maps SDK for academic deployment). Hopeline one-tap button.
Backend	ASP.NET MVC, C#, Visual Studio 2022	REST API; JWT authentication; BCrypt password hashing; POPIA consent module.
Database	Microsoft SQL Server (Developer Edition)	Nine tables in 3NF. Azure SQL for production deployment.
Email Service	Mailgun / SendGrid API	Registration confirmation and referral notification emails.
Maps — Website	Google Maps JavaScript API	Interactive clinic locator with 16 site pins. API key required for production.
Maps — Mobile	OSMDroid (OpenStreetMap)	Used in academic build. Google Maps Android SDK available for production switch.
Authentication	JWT + BCrypt	HTTP-only cookie (website); in-memory token (mobile).
Hosting — Current	GitHub Pages + CloudLabs	Website V1 on GitHub Pages. Full stack on CloudLabs academic environment.
Hosting — Production	Azure Web App + Azure SQL	Documented in Maintenance and Handover Guide for COSUP deployment.
Compliance	POPIA Act 4 of 2013	Consent on all forms; no sensitive health data stored; full user data rights.

4.5 System Reports — Revised

The following six system reports are generated by the COSUP platform for administrative and operational use:

Report Name	Requested By	Data Source	Output Format
Referral Submissions Report	COSUP Admin Staff	Referral, User tables	PDF download / Screen table
Site Listing Report	Admin / Public	Site, SiteService, Service tables	PDF download / Printable web page
Hopeline Usage Report	COSUP Management	HopelineLog table	Screen chart + CSV export
Registered Users Report	COSUP Admin	User table	Screen table (admin panel)
Admin Activity Audit Report	COSUP IT / Management	AuditLog, User tables	Screen table + PDF download
Services Directory	Public — always visible	Service table	Screen display (website + mobile)

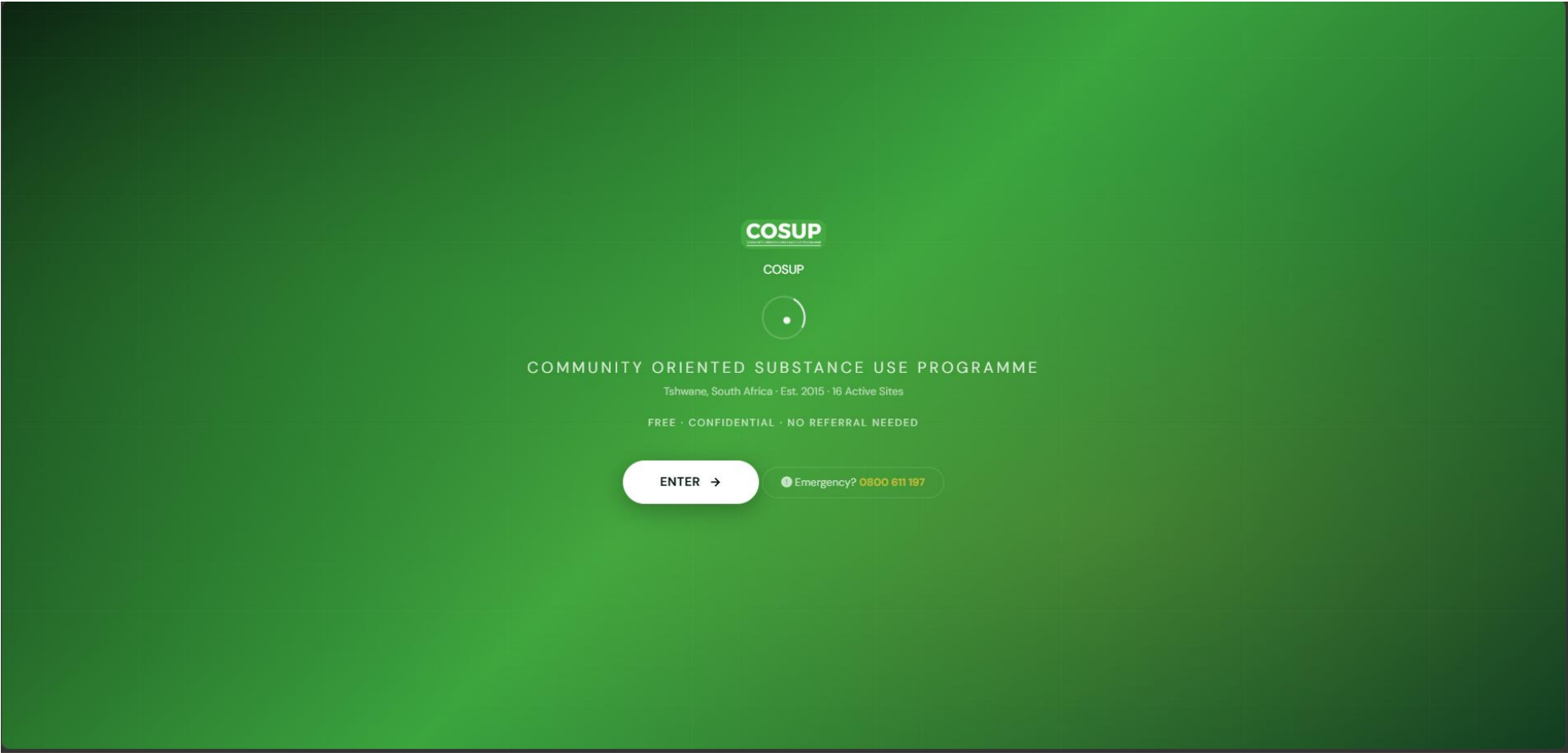
5. Prototype Screenshots

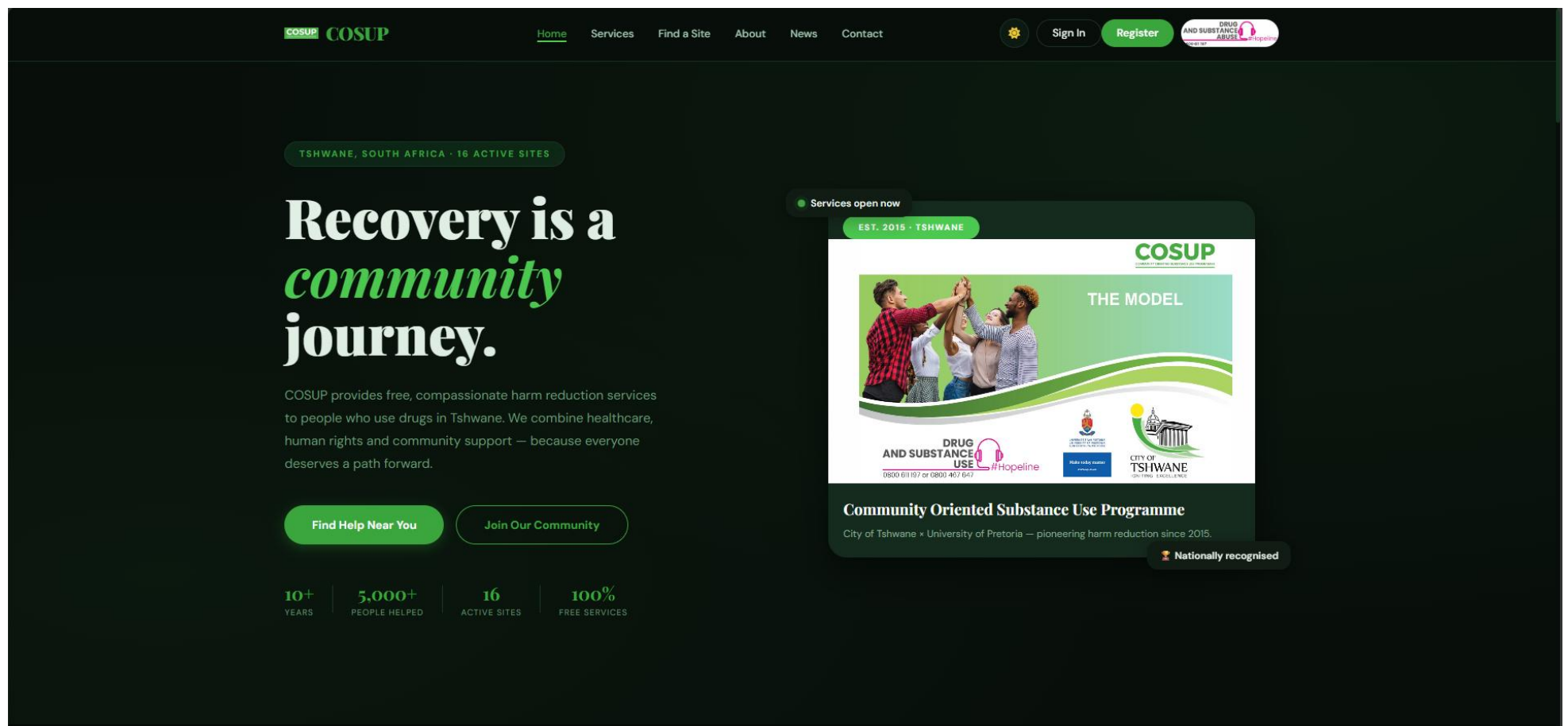
5.1 Website Prototype — Overview

The following section documents the key screens of the COSUP website prototype (Website Version 1), which was deployed to GitHub Pages and is accessible at the working link provided in Section 6 of this report. All screenshots reflect the fully functional delivered system as at June 2026.

5.1.1 Home Page — Dark Mode

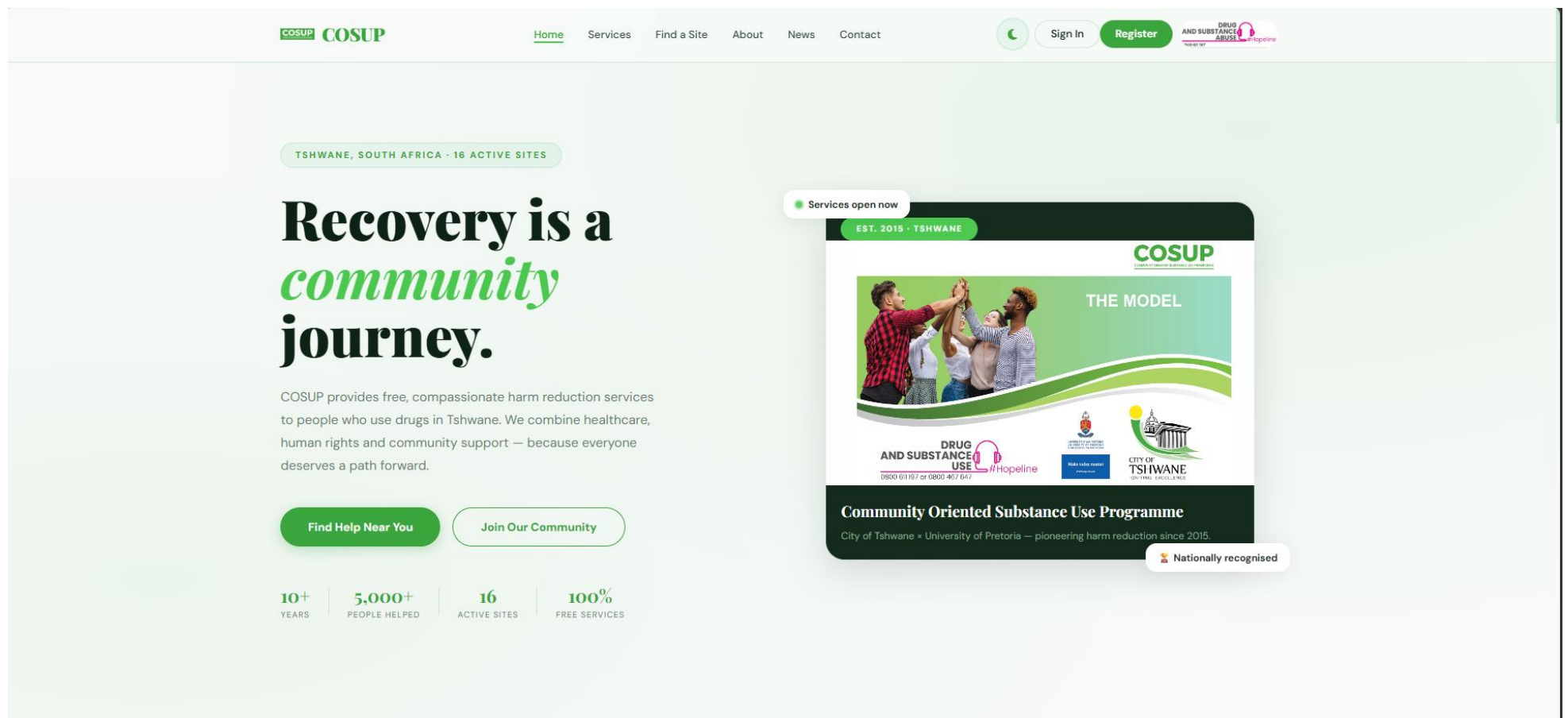
The homepage opens in dark mode by default. It features the hero headline ('Recovery is a community journey.'), a live-status indicator ('Services available now'), an impact statistics bar (10+ Years | 5 000+ People Supported | 16 Active Sites | 100% Free), service tiles, and the Hopeline box displaying 0800 611 197. The navigation bar includes Home, Services, Find a Site, About, and Contact, with Sign In, Register, and the Hopeline call-to-action buttons on the right.





5.1.2 Home Page — Light Mode

The light mode toggle switches the interface to a clean off-white background with the same content and layout. The mode toggle is accessible via the sun/moon icon in the navigation bar and persists for the user's session.



5.1.3 Services Page

The Services page displays all six COSUP services in a responsive card grid: Opioid Agonist Therapy, Needle & Syringe Programme, Psychosocial Support, Skills Development & Vocational Training, HIV & HCV Prevention, and Community Integration. Each card includes the service name, a descriptive icon image, and a full description.

COSUPCOSUP

[Home](#)[Services](#)[Find a Site](#)[About](#)[News](#)[Contact](#)

[Sign In](#)[Register](#)

WHAT WE OFFER

Comprehensive care for every step of your journey

All COSUP services are free of charge. No referral needed. Walk in or contact us directly at any of our 16 sites.

Opioid Agonist Therapy

Evidence-based medication-assisted treatment to reduce dependence and support long-term recovery in a safe, non-judgemental environment.

TREATMENT

Needle & Syringe Programme

Free, sterile equipment to reduce the risk of HIV, Hepatitis C and other blood-borne infections for people who inject drugs.

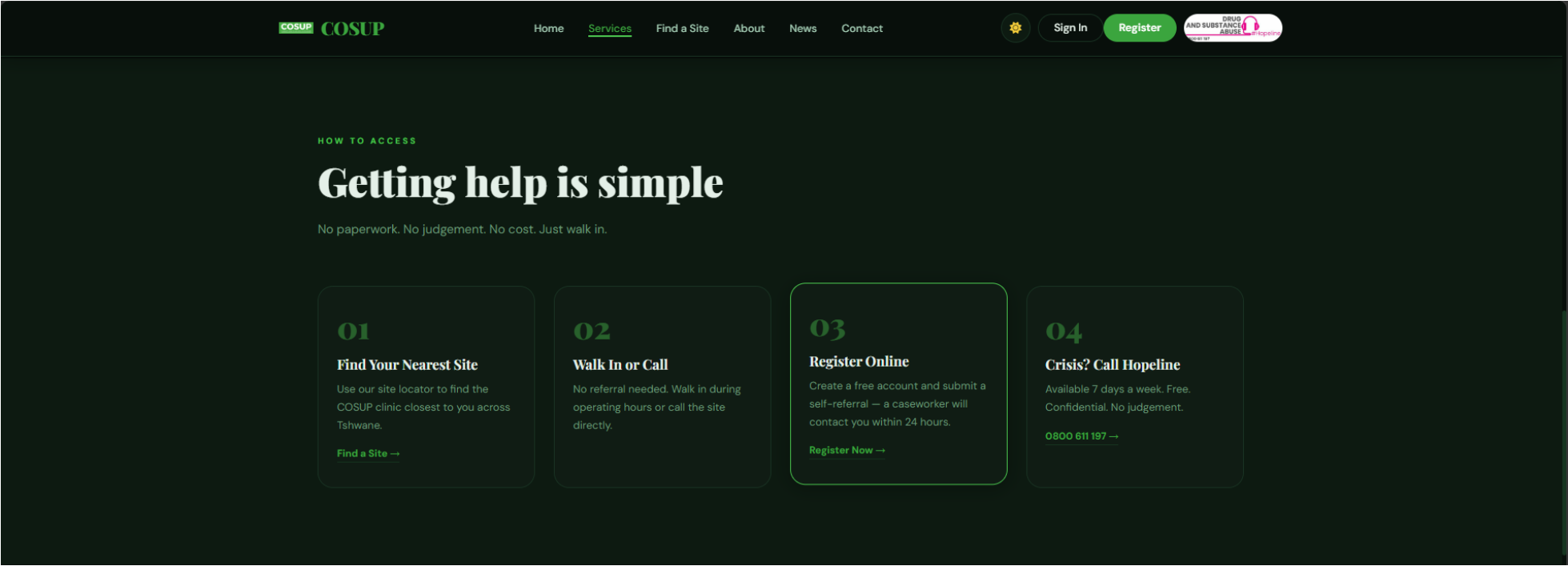
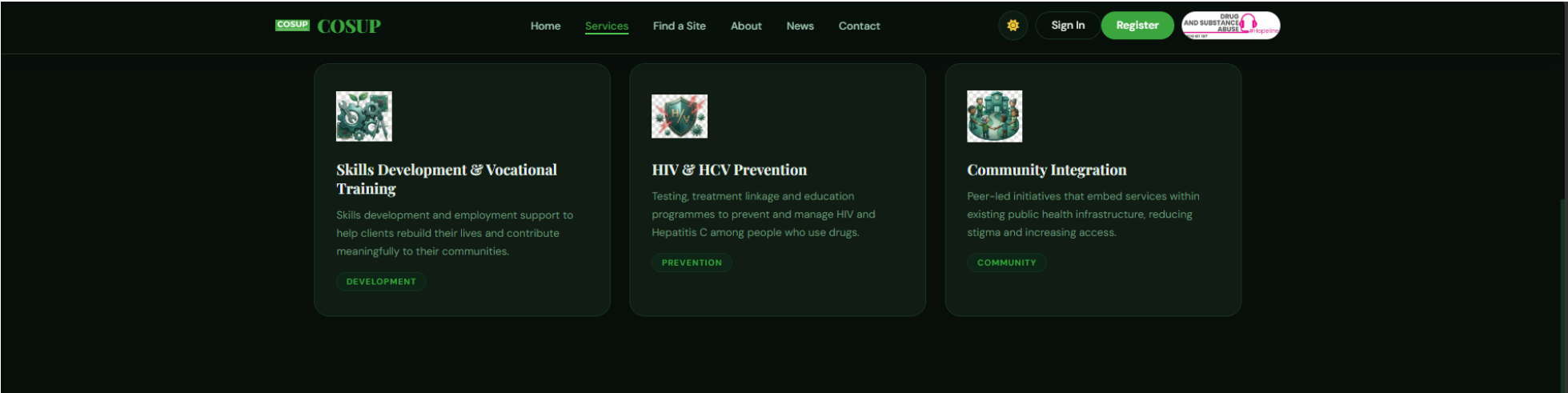
PREVENTION

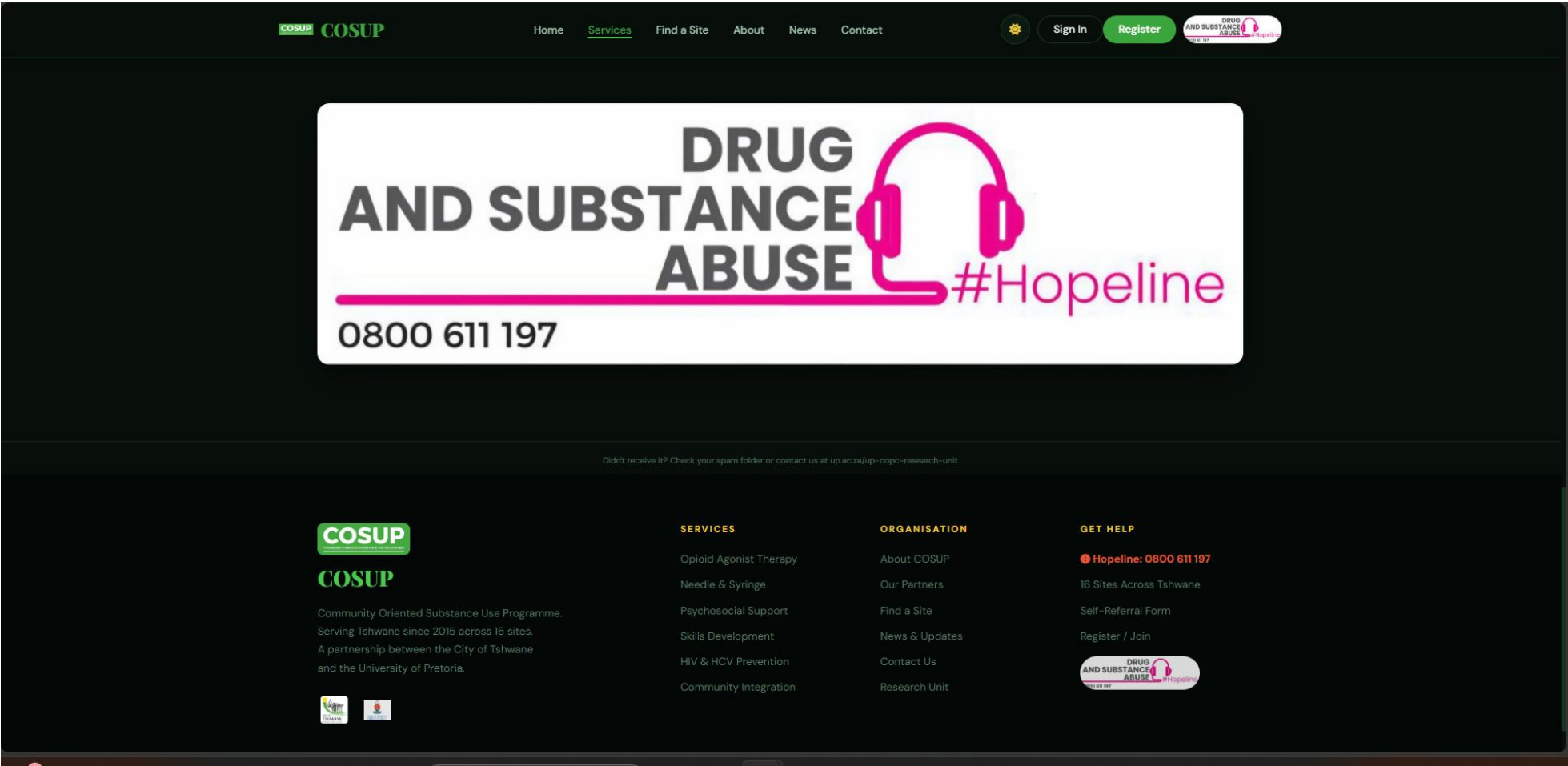
Psychosocial Support

Counselling, peer support groups and mental health services to address the emotional and social aspects of substance use.

SUPPORT

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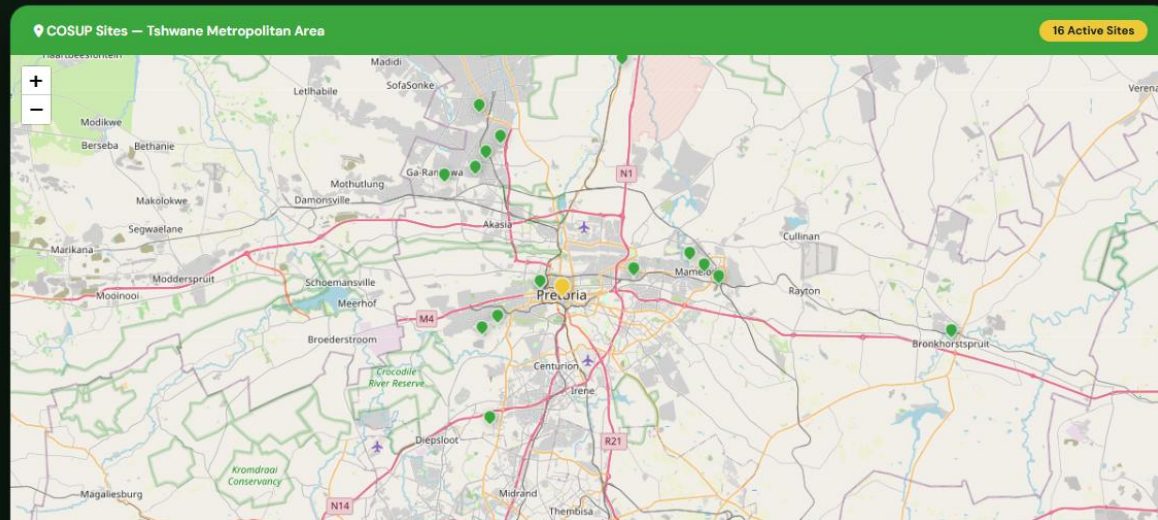


5.1.4 Find a Site Page

The Find a Site page displays a Google Maps-integrated interactive map with all 16 COSUP site pins plotted across the Tshwane metropolitan area. Below the map, a searchable grid lists all 16 sites with their address, operating hours, phone number, and service tags. Users can search by area name, site name, or service type.

16 Sites Across Tshwane

Click any pin on the map to get directions. All services are free. No referral needed.



COSUP

HomeServicesFind a SiteAboutNewsContact

Sign In

Register

DRUG AND SUBSTANCE ABUSE Support Programme

COSUP Sites — Tshwane Metropolitan Area

16 Active Sites

Click any pin → get driving, walking or transit directions. Gold pin = Main Centre.

01

COSUP Sediba Hope — Bosman St
Main Centre

📍 173 Bosman Street, Pretoria CBD

🕒 Mon–Fri 08:00–16:00

🏠 Community Integration, HIV & HCV Prevention, Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support, Skills Development & Vocational Training

Drive

Walk

02

COSUP Atteridgeville Clinic

📍 1A Mareka Street, Atteridgeville

🕒 Mon–Fri 08:00–16:00

🏠 HIV & HCV Prevention, Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support

☎️ 082 858 2450

Drive

Walk

03

COSUP Bronkhorstspuit

📍 Zithobeni Clinic, Bronkhorstspuit

🕒 Tue & Thur 09:30–14:00

🏠 HIV & HCV Prevention, Needle & Syringe Programme

☎️ 076 560 1389

Drive

Walk

04

COSUP Daspoort Poli Clinic

📍 Cnr Market & Camp Street, Pretoria West

🕒 Mon–Fri 08:00–16:00

🏠 Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support

☎️ 082 857 0922

Drive

Walk

COSUP

COSUP

Home

Services

Find a Site

About

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Register

01

COSUP Sediba Hope — Bosman St
Main Centre

173 Bosman Street, Pretoria CBD

Mon–Fri 08:00–16:00

Community Integration, HIV & HCV Prevention, Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support, Skills Development & Vocational Training

082 858 4304

Drive

Walk

Transit

02

COSUP Atteridgeville Clinic

1A Mareka Street, Atteridgeville

Mon–Fri 08:00–16:00

HIV & HCV Prevention, Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support

082 858 2450

Drive

Walk

Transit

03

COSUP Bronkhorstspuit

Zithobeni Clinic, Bronkhorstspuit

Tue & Thur 09:30–14:00

HIV & HCV Prevention, Needle & Syringe Programme

076 560 1389

Drive

Walk

Transit

04

COSUP Daspoort Poli Clinic

Cnr Market & Camp Street, Pretoria West

Mon–Fri 08:00–16:00

Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support

082 857 0922

Drive

Walk

Transit

05

COSUP Eersterust Community Health Centre

Cnr PS Fourie Drive & Hans Coovardale Road

Mon–Fri 08:00–16:00

HIV & HCV Prevention, Needle & Syringe Programme, Opioid Agonist Therapy

082 841 6038

Drive

Walk

Transit

06

COSUP Garankuwa

Readlegle Centre, Stand 05030

Mon–Fri 08:00–16:00

HIV & HCV Prevention, Psychosocial Support

079 321 4152

Drive

Walk

Transit

07

COSUP Hammanskraal

Mandisa Shiceka Clinic, Old Warmbaths Road

Mon & Fri 08:00–16:00

Opioid Agonist Therapy, Psychosocial Support

086 300 8338

Drive

Walk

Transit

08

COSUP Laudium Community

405 Bengal Street, Laudium

Mon–Fri 08:00–16:00

HIV & HCV Prevention, Opioid Agonist Therapy, Psychosocial Support

081 725 2462

Drive

Walk

Transit

09

COSUP Mamelodi Ikageng

21882 Molokoloko Circle, Mamelodi

Mon–Fri 08:00–16:00

HIV & HCV Prevention, Psychosocial Support

076 560 1389

Drive

Walk

Transit

10

COSUP Mamelodi Lusaka

Cnr Millenysane & Ratshwene Street

Mon–Fri 08:00–16:00

Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support

082 858 2553

Drive

Walk

Transit

11

COSUP Mamelodi Regional Hospital

Cnr Serapeng & Tsarnaya Road, Mamelodi

Mon–Fri 08:00–16:00

HIV & HCV Prevention, Needle & Syringe Programme, Opioid Agonist Therapy, Psychosocial Support

066 472 5580

Drive

Walk

Transit

12

COSUP Olievenhoutbosch

Cnr Imbovane & Imbongolo Street

Mon–Fri 08:00–16:00

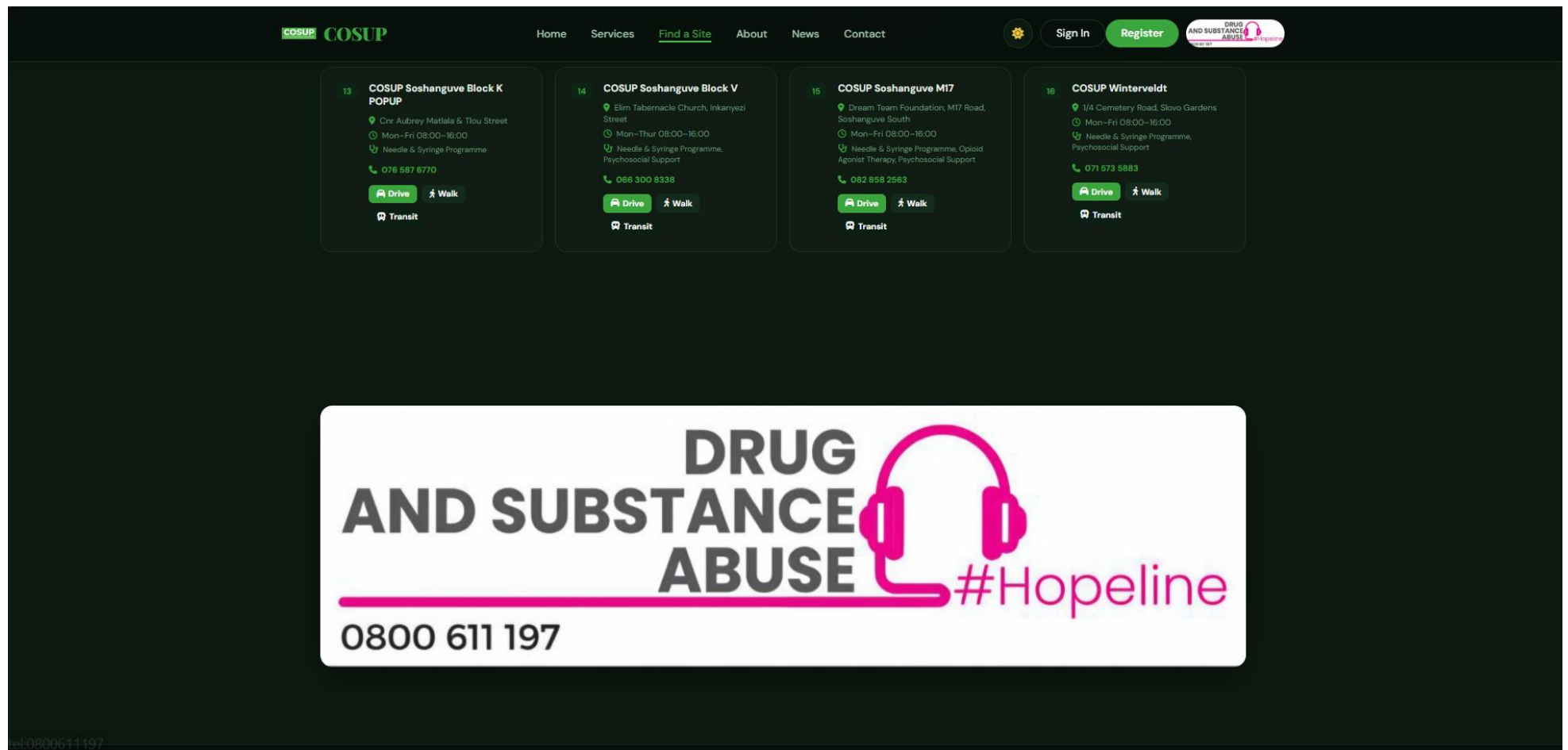
HIV & HCV Prevention, Needle & Syringe Programme

066 472 5740

Drive

Walk

Transit



5.1.5 About Us Page

The About Us page presents COSUP's story under the headline 'A decade of compassionate, evidence-based care', with partnership cards for the City of Tshwane (primary funder and operational partner) and the University of Pretoria, Department of Family Medicine (research partner). An image from COSUP field operations is featured.

COSUP

COSUP

Home

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DRUG AND SUBSTANCE ABUSE

Support

OUR STORY

A decade of compassionate, evidence-based care

Serving Tshwane since 2015. Nationally and internationally recognised. 100% free. No referral needed.

WHO WE ARE

Rights-based harm reduction for Tshwane

COSUP was established in 2015 in response to the growing limitations of abstinence-based and punitive approaches to substance use in Tshwane. We believe that people who use drugs deserve dignified, rights-based healthcare — regardless of their circumstances.

Through a formal partnership between the City of Tshwane and the University of Pretoria's Department of Family Medicine, COSUP has become a nationally and internationally recognised model for community-oriented harm reduction.





City of Tshwane
Primary funder and operational partner since 2015

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COSUPCOSUP

HomeServicesFind a SiteAboutNewsContact

⚙️

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DRUG AND SUBSTANCE ABUSE PROGRAM

TERMS

Our approach is grounded in evidence, guided by human rights, and delivered by a dedicated team of healthcare workers, social workers, and peer counsellors embedded within existing public health infrastructure.

Get Involved

University of Pretoria

Department of Family Medicine - Research partner

10+

Years of Operation

5,000+

People Supported

100%

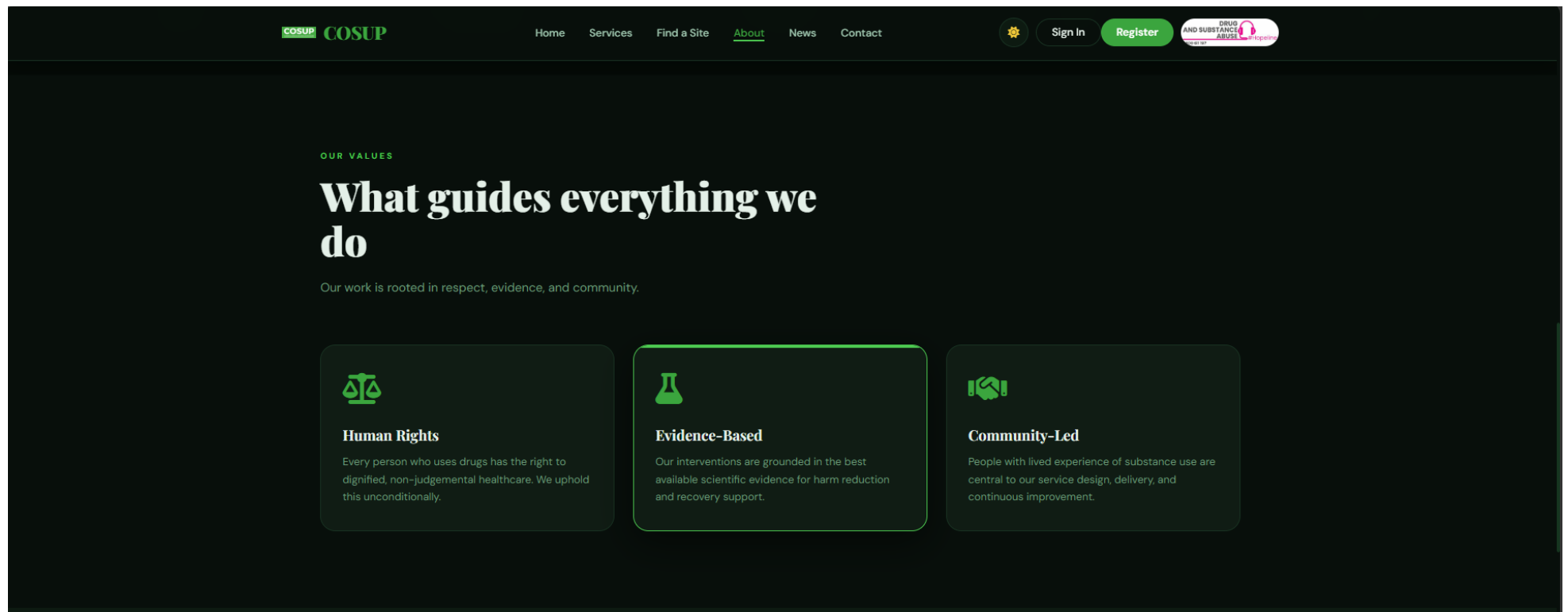
Free Services

16

Active Sites

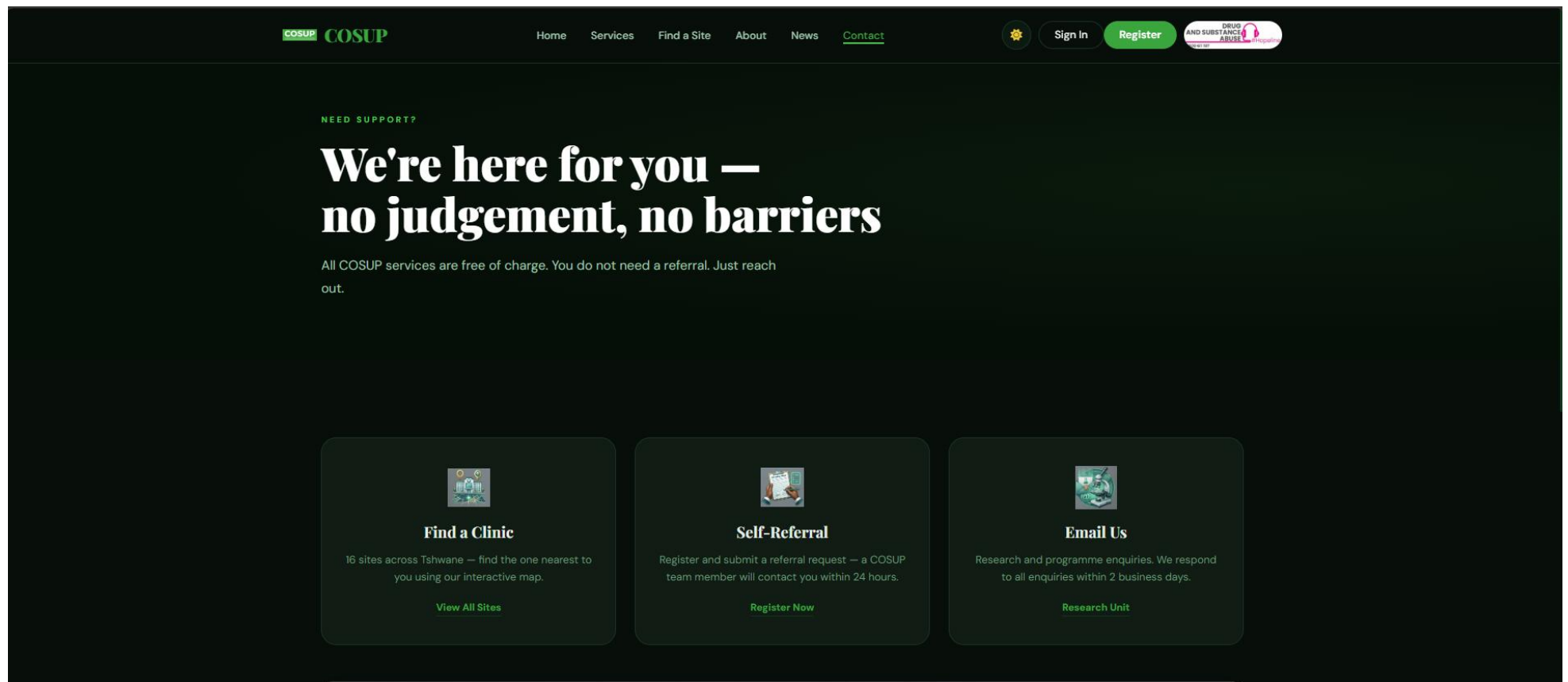
2

Institutional Partners



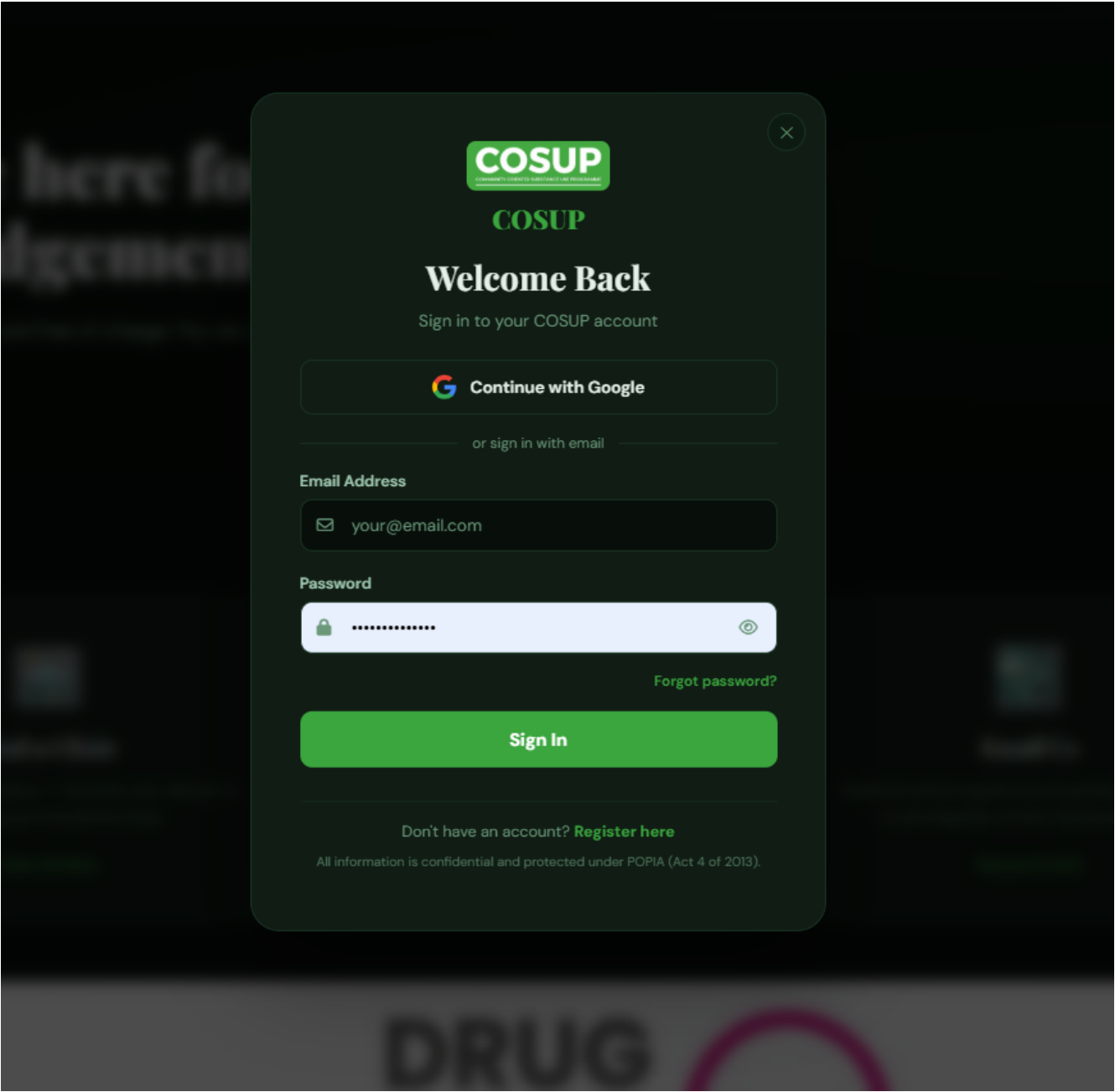
5.1.6 Contact Page

The Contact page presents three clear options under 'We're here for you — no judgement, no barriers': Find a Clinic, Call or WhatsApp, and Self-Referral. The Hopeline banner prominently displays 0800 611 197 in an SOS-styled orange container, available 7 days a week, free, confidential, and without judgement.



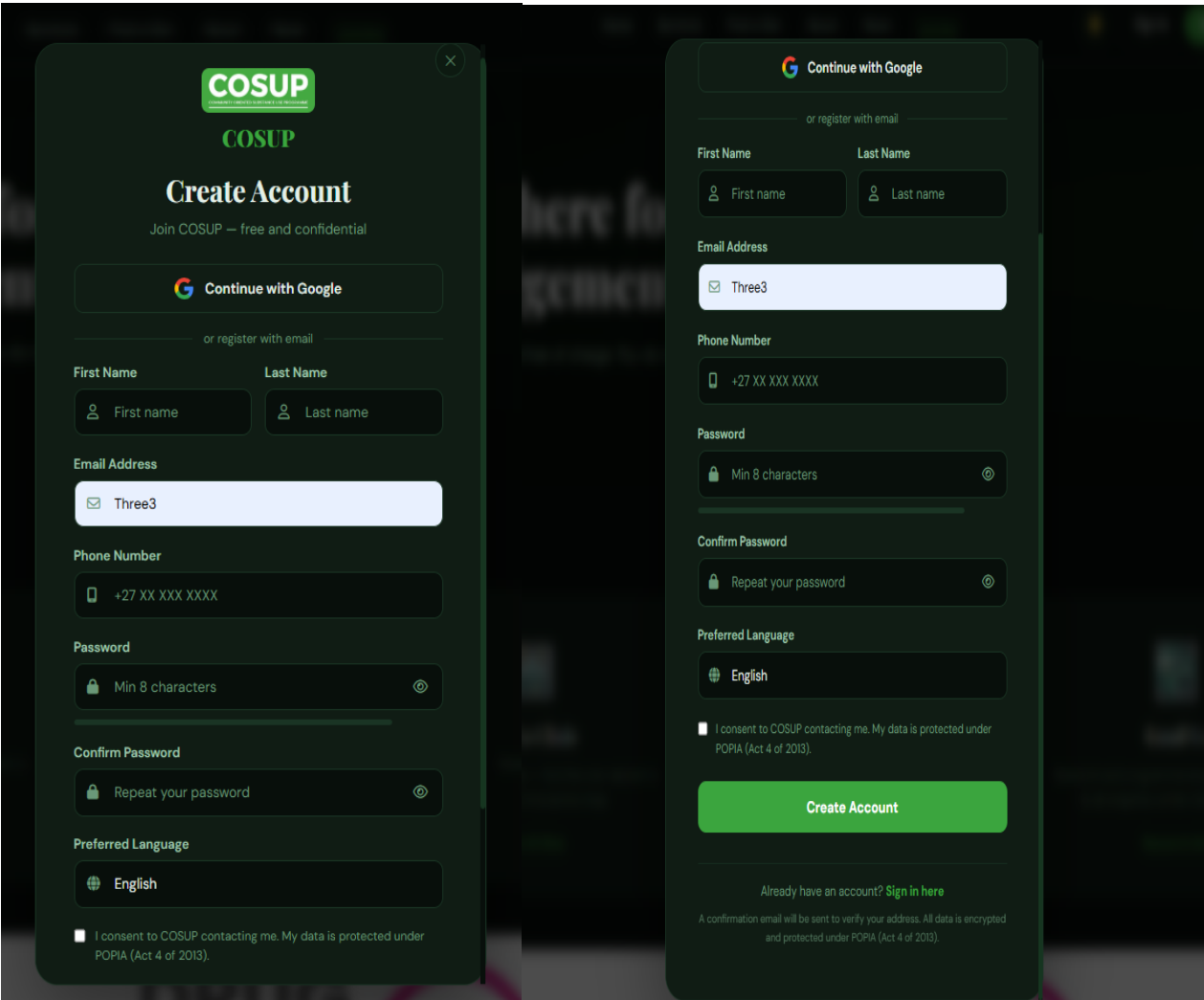
5.1.7 Sign In Page

The Sign In modal features the COSUP logo, email address and password fields, and a 'Sign In' button. A link to the registration form is provided for new users. A POPIA compliance notice is displayed at the bottom of the modal.



5.1.8 Register Page

The Register modal collects First Name, Last Name, Email Address, Phone Number, Password (minimum 8 characters), and Preferred Language (dropdown with all 7 supported languages). A mandatory POPIA consent checkbox reads: 'I consent to COSUP contacting me and understand my data is protected under POPIA.' A confirmation email is sent on successful registration.



5.2 Mobile Application Prototype — Overview

The Android mobile application was developed in Kotlin using Android Studio. The following screens reflect the fully functional delivered application as at June 2026.

5.2.1 Home Screen

The home screen opens with the COSUP branding, a personalised welcome, and the Hopeline SOS button prominently displayed in an orange banner ('Hopeline — 0800 611 197 — Tap to call our support line immediately'). Below are four service tiles (Opioid Therapy, Needle & Syringe, Psychosocial, Skills Development) and a bottom navigation bar (Home | Find Help | Services | Profile).

[SCREENSHOT: Mobile App Home Screen — Insert screenshot here]

5.2.2 Find Help Screen — 16 Sites Map

The Find Help screen presents an interactive map with all 16 COSUP site pins plotted across Tshwane (implemented using OSMDroid). Below the map, a scrollable list displays all 16 sites with name, address, hours, and service tags (OAT, NSP, Counselling, etc.). The Sediba Hope — Bosman St site is marked as the Main Centre.

[SCREENSHOT: Mobile App Find Help Screen — Insert screenshot here]

5.2.3 Mobile App — Light Mode

The mobile application supports both dark and light themes. The light mode version retains full functionality with COSUP green branding on a white background.

[SCREENSHOT: Mobile App Light Mode — Insert screenshot here]

6. Working System Links

The following links provide direct access to the working deployed components of the COSUP digital platform as at June 2026:

Component	Link / Access Detail
UI/UX Prototype (Figma/HTML)	https://tmosetlha.github.io/cosup-prototype/
Website Version 1 (Live)	https://tmosetlha.github.io/COSUP_WEBSITE_VERSION_1/
GitHub Repository	https://github.com/tmosetlha — Source code for all project deliverables
Android APK / Demo	Available via GitHub Releases — cosupv2 repository (com.example.cosupv2)
Admin Demo Credentials	Username: admin@cosup.org.za Password: [provided to evaluator separately for security]

Note: The full ASP.NET MVC backend and Microsoft SQL Server database are hosted on the CloudLabs academic environment. CloudLabs may be deallocated after 22 June 2026 as per the module coordinator's notice. All source code, database scripts, and screenshots have been backed up to GitHub and OneDrive as instructed.

6.1 CRUD Demonstration

Both the website and the mobile application are fully connected to a database and demonstrate all four CRUD operations as required by the final submission specification:

Operation	Feature	Description
CREATE	User Registration / Self-Referral / News Publish	New user records created on registration. Referral submissions inserted into Referral table. Admin publishes new NewsItem records.
READ	Clinic Locator / Services / News Feed / User Login	All site, service, and news data retrieved from the database and displayed on the website and mobile app. User credentials verified on login.

UPDATE	Admin — Update Clinic Hours / Referral Status	Authorised admin updates clinic operating hours and addresses via the admin panel. Referral status updated from 'pending' to 'contacted'. All updates logged in AuditLog.
DELETE	Admin — Remove Outdated News / User Account	Admin can unpublish or delete news items. Users can request deletion of their account in compliance with POPIA data deletion provisions. Deletions logged in AuditLog.

6.2 User Role Differentiation

The system implements two distinct user types as required by the final submission specification:

- Admin User — authenticated via role-based JWT. Access to clinic management, news publishing, referral review, and audit log. Admin-only routes return HTTP 401 for unauthenticated requests.
- Normal User (Registered Public) — authenticated via JWT. Access to personal profile, preferred language setting, and self-referral submission with POPIA consent. Cannot access admin functions.
- Anonymous Visitor — no authentication required. Full access to all public-facing content: clinic locator, services, news, Hopeline, About, Contact.

6.3 Input Validation

The system includes proper input validation across all data capture forms as required:

- Email fields: format validation (must contain @ and valid domain structure).
- Password fields: minimum 8 characters enforced; BCrypt-hashed before storage.
- Phone fields: accepts South African format (+27 or 0 prefix); optional on registration.
- POPIA consent checkbox: mandatory on both registration and self-referral forms; form cannot be submitted without ticking.
- Admin update forms: all Site fields validated server-side before database write; audit log entry created on every successful update.
- SQL injection prevention: all database interactions use parameterised queries via Entity Framework / ADO.NET.

7. PowerPoint Presentation Slides

7.1 Overview

The team prepared a formal PowerPoint presentation for the XISD5319 Final Submission. The presentation was designed to be delivered to both COSUP management and IIE Rosebank International academic evaluators. The full .pptx file is submitted alongside this report as a separate deliverable.

The slide deck follows the COSUP brand palette (dark green #1A7A1A, black #0D0D0D, white) and typography (Playfair Display for headings, DM Sans for body text), consistent with the website and mobile application design language.

7.2 Slide Structure

#	Slide Title	Key Content
1	Title Slide	COSUP Website & Mobile Application Task 3 Final Submission The 5 Watermelons XISD5319 June 2026
2	Team Introduction	Team member names, roles, and student numbers. 'The 5 Watermelons — Built in Code.'
3	The Problem	COSUP's 10-year digital gap. Three critical operational problems: discoverability, fragmented information, limited reach.
4	Our Solution	Two integrated platforms: responsive website and Android mobile application. Dual audience: general public and people in crisis.
5	Key Features — Website	Clinic locator (16 sites), multilingual toggle (7 languages), Hopeline box, self-referral, admin portal, dark/light mode.
6	Key Features — Mobile App	OSMDroid map with 16 pins, one-tap Hopeline SOS button, services directory, news feed, POPIA-compliant registration.
7	Live Demo	Live walkthrough of Website V1 and Android APK. Demonstrates CREATE, READ, UPDATE, DELETE operations.

8	Database & Architecture	Three-tier architecture diagram. Nine-table SQL Server schema. POPIA compliance summary.
9	Testing Summary	Functional testing results. Cross-browser and cross-device testing. UAT feedback from COSUP.
10	Challenges & Solutions	Multilingual complexity resolved via i18n JSON bundles. COSUP response delays managed independently. OSMDroid adopted for maps.
11	Project Impact	Digital reach for 5 000+ beneficiaries. First-ever COSUP digital presence. Estimated commercial value R265 000–R505 000.
12	Handover Summary	Source code on GitHub. Database scripts included. Maintenance and Handover Guide prepared for COSUP.
13	Closing Slide	'Recovery is a community journey.' COSUP Hopeline: 0800 611 197 The 5 Watermelons IIE Rosebank International 2026

Note: The full PowerPoint presentation (.pptx) is submitted as a separate file alongside this report, as required by the XISD5319 Final Submission guidelines.